

Module 08: Planning — Neural Circuits for Prospection

Course: Active Inference 101 | **Unit:** Computational Neuroscience | **Audience:** First-semester undergraduates

Learning Objectives

1. Describe the **default mode network (DMN)** and its role in imagination, planning, and self-referential thought.
2. Explain how the **hippocampus** supports future thinking through constructive memory.
3. Identify the neural basis of **goal-directed vs. habitual behavior** through prefrontal-striatal interactions.

Introduction

Planning — thinking about the future — is one of the most remarkable things brains do. This module reveals that the neural systems for remembering the past, imagining the future, and daydreaming are largely the same. The brain doesn't just store memories — it uses them as building blocks to construct possible futures.

Key Concepts

1. The Default Mode Network (DMN)

The **DMN** is a set of brain regions active when you're *not* focused on the external world:

- **Medial prefrontal cortex (mPFC):** Self-referential thinking, social reasoning
- **Posterior cingulate cortex (PCC):** Memory retrieval, personal significance
- **Angular gyrus:** Semantic memory, conceptual knowledge
- **Hippocampus:** Memory construction and recombination

The DMN activates during:

- Daydreaming and mind-wandering
- Imagining future scenarios
- Remembering past events
- Thinking about others (Theory of Mind)

In Active Inference, the DMN runs the generative model in "offline" mode — simulating possible futures without acting.

2. Hippocampal Prospection — Constructing Futures

The hippocampus doesn't just replay memories — it **recombines** them to construct novel scenarios:

- Memory: "Last time I went to the beach, it was sunny and warm"
- Imagination: "If I go to the beach *tomorrow*, and it's raining, I should bring an umbrella"

Patients with hippocampal damage have impaired imagination — they can't construct rich future scenarios. This confirms that **memory and imagination share neural machinery**.

In Active Inference, this is the brain running its generative model forward in time, evaluating different policies.

3. Prefrontal-Striatal Interactions: Goals vs. Habits

Two neural systems compete for behavioral control:

Goal-directed system (prefrontal cortex → caudate nucleus):

- Evaluates actions based on their expected outcomes
- Flexible but computationally expensive
- Dominant when situations are novel or outcomes change

Habitual system (sensorimotor cortex → putamen):

- Selects actions based on learned stimulus-response associations
- Fast and automatic but inflexible

- Dominant when situations are familiar and well-practiced

Active Inference predicts a smooth transition: as prediction errors decrease with practice, behavior shifts from goal-directed (deliberate EFE evaluation) to habitual (strong policy priors).

4. Time Cells and Temporal Coding

The brain represents time through **time cells** — hippocampal neurons that fire at specific moments in a temporal sequence:

- Different cells fire at $t=1s$, $t=2s$, $t=3s$, etc. during a waiting period
- This creates a neural "timeline" that supports temporal planning
- Combined with place cells, they form a space-time map for navigation and planning

5. Disruptions of Planning

Condition	Neural Basis	Planning Deficit
Frontal lobe damage	PFC dysfunction	Impaired future thinking, impulsivity
Hippocampal amnesia	Hippocampal damage	Cannot imagine future scenes
Depression	Reduced DMN flexibility	Repetitive negative future thinking (rumination)
Addiction	Hijacked dopaminergic evaluation	Over-valuation of immediate rewards, ignoring future costs

Summary

Planning depends on a network including the DMN, hippocampus, and prefrontal-striatal circuits. The DMN runs the generative model offline, the hippocampus constructs novel future scenarios from memory fragments, and prefrontal-striatal interactions balance goal-directed

and habitual control. Time cells provide the temporal scaffolding for planning across timescales.

Further Reading

- Buckner, R. L. & Carroll, D. C. (2007). Self-projection and the brain. *Trends in Cognitive Sciences*, 11(2), 49-57.
- Hassabis, D. & Maguire, E. A. (2007). Deconstructing episodic memory with construction. *Trends in Cognitive Sciences*, 11(7), 299-306.
- Daw, N. D., Niv, Y., & Dayan, P. (2005). Uncertainty-based competition between prefrontal and dorsolateral striatal systems for behavioral control. *Nature Neuroscience*, 8(12), 1704-1711.